Supplementary Material for the Manuscript "GBFRVFL: Granular-Ball Computing-Based Fuzzy Random Vector Functional Link Network"

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I. Results and Discussion on UCI and KEEL Datasets in Noisy Environments

The UCI and KEEL datasets considered in this subsection represent realistic learning scenarios in which training data are frequently corrupted by label noise. Evaluating model robustness under such adverse conditions is therefore essential for assessing the reliability and practical applicability of learning algorithms. To this end, controlled levels of label noise are intentionally introduced into five representative datasets, namely cleve, conn_bench_sonar_mines_rocks, ecoli-0-1-4-6_vs_5, haberman_survival, and ionosphere, with noise ratios set to 5%, 10%, 20%, 30%, and 40%. The performance of the proposed F-GBRVFL and SDAP-GBRVFL models is compared against the baseline models. The corresponding classification Acc are reported in Table I. From the results, it is evident that both proposed models demonstrate strong resilience to increasing levels of label noise across all datasets. In particular, the SDAP-GBRVFL model consistently achieves the highest or second-highest Acc in most noise settings, highlighting the effectiveness of the SDAP membership mechanism in suppressing the influence of mislabeled samples. On the cleve dataset, SDAP-GBRVFL attains the best average Acc among all competing models, maintaining stable performance even when the noise level reaches 40%. This indicates that the adaptive membership weighting successfully mitigates the adverse impact of label corruption. Similar trends can be observed on the conn_bench_sonar_mines_rocks dataset, where both F-GBRVFL and SDAP-GBRVFL outperform conventional RVFL-based models across nearly all noise levels. On the ecoli-0-1-4-6_vs_5 dataset, the proposed models consistently deliver strong performance, with SDAP-GBRVFL achieving the best overall average Acc. This behavior confirms that granular-ball abstraction, combined with adaptive fuzzy membership assignment, effectively reduces the influence of corrupted labels. On the ionosphere dataset, both proposed models maintain stable Acc as noise increases, outperforming most baseline models at higher noise levels and demonstrating improved generalization under noisy conditions. When considering the overall average Acc across all datasets and noise ratios, SDAP-GBRVFL achieves the best performance, followed closely by F-GBRVFL. These results clearly indicate that incorporating fuzzy and statistical density-adaptive Pythagorean memberships into the RVFL framework substantially enhances robustness to label noise.

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II. Sensitivity Analysis

This subsection examines the robustness and stability of the proposed F-GBRVFL and SDAP-GBRVFL models by analyzing their sensitivity to key hyperparameters. The study evaluates how changes in the regularization parameter, choice of activation function, and model capacity affect classification performance across multiple datasets.

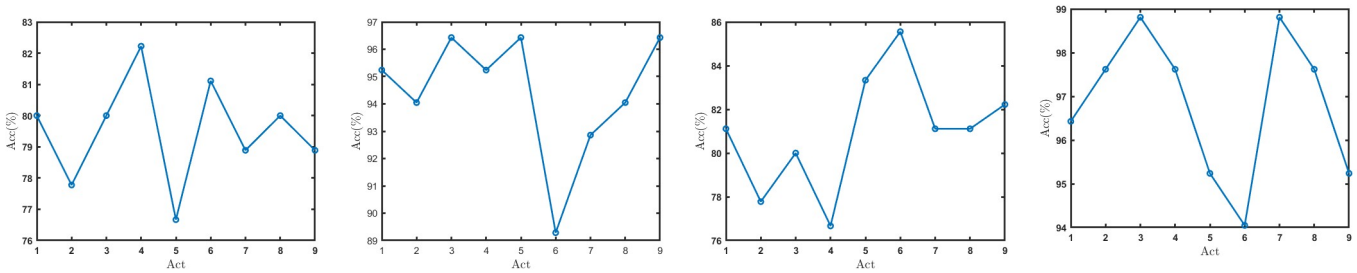
A. Sensitivity analysis of the activation function (Act)

Figure 1 illustrates the influence of different activation functions on the classification Acc of the proposed F-GBRVFL and SDAP-GBRVFL models across two representative datasets, namely cleve and ecoli-0-1-4-6_vs_5. Nine commonly used activation functions are evaluated to analyze the robustness of the proposed models with respect to activation choice. As shown in Fig. 1a and Fig. 1b, F-GBRVFL exhibits relatively stable performance across most activation functions, with peak Acc achieved using mid-range nonlinear activations, indicating that the fuzzy membership mechanism effectively alleviates sensitivity to activation selection. However, certain activation functions lead to noticeable performance drops, particularly in more complex or imbalanced datasets, reflecting the inherent variability of random hidden mappings. In contrast, Fig. 1c

TABLE I: Performance comparison of the proposed F-GBRVFL and SDAP-GBRVFL models along with the baseline models for UCI and KEEL datasets with contaminated label noise.

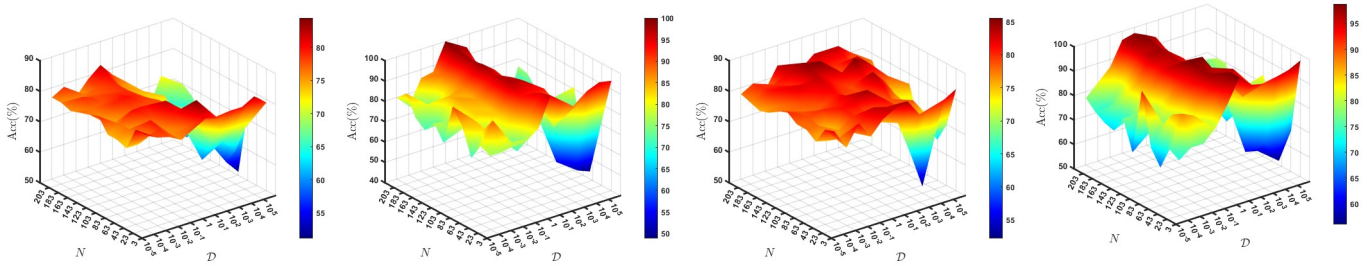
Dataset	Noise	RVFL [1]	ELM [2]	GB-RVFL [3]	GE-GB-RVFL [3]	CRVFL [4]	ACRVFL [4]	F-GBRVFL†	SDAP-GBRVFL†
cleve	5%	75.11	70.78	70	73.33	70.56	70.56	78.89	80
	10%	82.22	80	81.11	61.11	65.56	65.56	82.22	81.11
	20%	80	75.11	75.56	66.67	65.56	64.89	76.67	77.78
	30%	72.22	75.56	63.33	57.78	62.36	62.36	76.67	74.44
	40%	57.78	58.89	66.67	52.22	45.56	45.56	53.33	57.78
Average Acc		73.47	72.07	71.33	62.22	61.92	61.79	73.56	74.22
conn_bench_sonar_mines_rocks	5%	76.19	63.49	71.43	76.19	66.67	68.25	76.19	79.37
	10%	70.19	70.78	69.84	73.02	68.25	65.08	78.38	80.95
	20%	70.95	76.19	74.6	77.78	69.84	69.84	74.6	72.25
	30%	71.43	73.02	71.43	76.19	52.38	52.38	72.21	73.73
	40%	66.67	52.38	69.84	69.84	41.27	42.86	70.44	71.79
Average Acc		71.09	67.17	71.43	74.6	59.68	59.68	74.36	75.62
ecoli-0-1-4-6_vs_5	5%	90	90	83.33	94.05	90.95	90.95	96.43	98.81
	10%	98.81	98.81	98.81	94.05	95.95	95.95	95.24	94.05
	20%	98.81	98.81	98.81	100	95.95	95.95	98.67	98.05
	30%	90.86	90.05	94.05	96.43	55.95	55.95	91.67	94.05
	40%	64.05	61.67	65.48	58.33	45.95	45.95	64.76	69.52
Average Acc		88.51	87.87	88.1	88.57	76.95	76.95	89.35	90.9
haberman_survival	5%	72.17	72.09	75	77.17	83.7	82.61	67.39	80.43
	10%	77.17	75.35	76.09	77.17	79.35	73.91	81.52	85.22
	20%	79.35	77.17	75.35	77.17	75	82.61	81.52	80.43
	30%	51.09	59.35	76.09	61.96	82.61	80.43	80.74	81.52
	40%	55.43	52.17	56.52	53.91	53.26	58.7	60.43	62.91
Average Acc		67.04	67.23	71.81	69.48	74.78	75.65	74.32	78.1
ionosphere	5%	85.68	85.74	86.79	83.02	83.02	85.85	86.79	86.79
	10%	86.74	82.08	86.79	87.74	83.96	83.96	88.68	86.79
	20%	85.58	83.96	82.74	81.13	83.02	82.08	83.96	83.02
	30%	83.96	77.36	70.75	74.34	77.92	73.58	78.49	78.87
	40%	55.6	53.77	58.49	55.66	47.55	40.81	59.89	63.21
Average Acc		79.51	76.58	77.11	76.38	75.09	73.26	79.56	79.74
Overall Average Acc		75.92	74.18	75.96	74.25	69.68	69.47	<u>78.23</u>	**79.72**

The proposed model is denoted by †.
The top and second-best models in terms of ACC are denoted by boldface and underline, respectively.



(a) cleve (F-GBRVFL) (b) ecoli-0-1-4-6_vs_5 (F-GBRVFL) (c) cleve (SDAP-GBRVFL) (d) ecoli-0-1-4-6_vs_5 (SDAP-GBRVFL)

Fig. 1: The impact of changing the Act on the Acc values of the proposed F-GBRVFL and SDAP-GBRVFL models.



(a) cleve (F-GBRVFL) (b) ecoli-0-1-4-6_vs_5 (F-GBRVFL) (c) cleve (SDAP-GBRVFL) (d) ecoli-0-1-4-6_vs_5 (SDAP-GBRVFL)

Fig. 2: The impact of changing the parameters $\mathcal{D}$ and N on the Acc values of the proposed F-GBRVFL and SDAP-GBRVFL models.

and Fig. 1d demonstrate that SDAP-GBRVFL consistently attains higher Acc and exhibits reduced performance fluctuation across different activation functions. This improved stability can be attributed to the proposed SDAPM scheme, which adaptively adjusts membership and non-membership values based on statistical density, local sparsity, and granular-ball compactness.

B. Sensitivity analysis of hyperparameters $\mathcal{D}$ and N

Figure 2 illustrates the joint sensitivity analysis of the hyperparameters $\mathcal{D}$ and N, which respectively control the regularization strength and the number of hidden nodes in the proposed F-GBRVFL and SDAP-GBRVFL models. The parameters are varied over a wide range to assess model robustness and stability under different configurations. For the cleve dataset using F-GBRVFL (Fig. 2a), high and stable Acc values are observed when $\mathcal{D}$ assumes moderate values and N lies in an intermediate range. In contrast, extreme settings of either parameter, particularly very small $\mathcal{D}$ or excessively large N, lead to noticeable performance degradation, indicating overfitting or insufficient regularization. For the ecoli-0-1-4-6_vs_5 dataset under the F-GBRVFL framework (Fig. 2b), the model demonstrates strong sensitivity to $\mathcal{D}$, where moderate-to-large values consistently yield superior Acc across a broad range of N. In the case of SDAP-GBRVFL on the cleve dataset (Fig. 2c), the Acc surface appears smoother and more stable, with peak performance attained across a wider region of the parameter space. Similarly, for ecoli-0-1-4-6_vs_5 using SDAP-GBRVFL (Fig. 2d), the model achieves consistently high Acc over a broad range of $\mathcal{D}$ and N values, with only marginal degradation at extreme configurations.

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